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CHAPTER 1

Statistical Models, Goals, and Performance Criteria

1.1. Problem 1.1.1: Examples of Statistical Models

Give a formal statement of the following models identifying the probability laws of the data and the parameter space. State whether the model in question is parametric or nonparametric.

(a) A geologist measures the diameters of a large number n of pebbles in an old stream bed. Theoretical considerations lead him to believe that the logarithm of pebble diameter is normally distributed with mean μ and variance σ. He wishes to use his observations to obtain some information about μ and σ^2 but has in advance no knowledge of the magnitudes of the two parameters.

(b) A measuring instrument is being used to obtain n independent determinations of a physical constant μ. Suppose that the measuring instrument is known to be biased to the positive side by 0.1 units. Assume that the errors are otherwise identically distributed normal random variables with known variance.

(c) In part (b) suppose that the amount of bias is positive but unknown. Can you perceive any difficulties in making statements about μ for this model?

(d) The number of eggs laid by an insect follows a Poisson distribution with unknown mean λ. Once laid, each egg has an unknown chance p of hatching and the hatching of one egg is independent of the hatching of the others. An entomologist studies a set of n such insects observing both the number of eggs laid and the number of eggs hatching for each nest.

Answer. (a) Let sample space Ω be defined by

$$\Omega \equiv \{(P_1, \ldots, P_n) : P_i \text{ is a pebble of the old stream}\}.$$

Let $\mathbf{X} : \Omega \to \mathbb{R}^n$ be the random vector defined by

$$\mathbf{X} \equiv (X_1, \ldots, X_n),$$
$$X_i(P_1, \ldots, P_n) \equiv \log(\text{diameter of } P_i).$$

We assume that $\mathbf{X} \sim P_{\mu,\sigma^2}$, where P_{μ,σ^2} is a probability distribution on $\mathbb{R}^n$ according to which $X_1, \ldots, X_n$ are independent and identically distributed with a common $\mathcal{N}(\mu, \sigma^2)$ distribution. Let the parameter space Θ be defined by

$$\Theta \equiv \{(\mu, \sigma^2) : \mu \in \mathbb{R}, \sigma^2 > 0\}.$$

Finally, let model $\mathcal{P}$ be defined by

$$\mathcal{P} \equiv \{P_{\mu,\sigma^2} : (\mu, \sigma^2) \in \Theta\}.$$

$\mathcal{P}$ is parametric.

(b) Let sample space Ω be defined by

$$\Omega \equiv \{(D_1, \ldots, D_n) \in \mathbb{R}^n : D_i \text{ is a determination of the physical constant}\}.$$

Let $\mathbf{X} : \Omega \to \mathbb{R}^n$ be the random vector defined by

$$\mathbf{X} \equiv (X_1, \ldots, X_n),$$
$$X_i(D_1, \ldots, D_n) \equiv D_i.$$

Let $\mathbf{Y} = (Y_1, \ldots, Y_n) : \Omega \to \mathbb{R}^n$ be the random vector defined by

$$\mathbf{Y} = \mathbf{X} - \mu.$$

We assume that $\mathbf{Y} \sim P_{\sigma_0^2}$, where $P_{\sigma_0^2}$ is a probability distribution on $\mathbb{R}^n$ according to which $Y_1, \ldots, Y_n$ are independently and identically distributed with a common $\mathcal{N}(+0.1, \sigma_0^2)$ distribution, where σ_0^2 is the known variance. Therefore

$$X_i \sim \mathcal{N}(\mu + 0.1, \sigma_0^2), \quad \forall i = 1, \ldots, n.$$

That is to say, $\mathbf{X} \sim P_{\mu+0.1}$, where $P_{\mu+0.1}$ is a probability distribution on $\mathbb{R}^n$ according to which $X_1, \ldots, X_n$ are independent and identically distributed with a common $\mathcal{N}(\mu + 0.1, \sigma_0^2)$ distribution. Let the parameter space Θ be defined by

$$\Theta \equiv \{\mu + 0.1 : \mu + 0.1 \in \mathbb{R}\}.$$

Finally, let model $\mathcal{P}$ be defined by

$$\mathcal{P} \equiv \{P_{\mu+0.1} : \mu + 0.1 \in \Theta\}.$$

$\mathcal{P}$ is parametric.

(c) Yes. The difficulty is that we need one more parameter μ', and the parametrization becomes unidentifiable. Specifically, we need to assume that $\mathbf{Y} \sim P_{\mu', \sigma_0^2}$, where P_{μ', σ_0^2} is a probability distribution on $\mathbb{R}^n$ according to which $Y_1, \ldots, Y_n$ are independently and identically distributed with a common $\mathcal{N}(\mu', \sigma_0^2)$ distribution. Therefore

$$X_i \sim \mathcal{N}(\mu + \mu', \sigma_0^2), \quad \forall i = 1, \ldots, n.$$

That is to say, $\mathbf{X} \sim P_{\mu, \mu'}$, where $P_{\mu, \mu'}$ is a probability distribution on $\mathbb{R}^n$ according to which $X_1, \ldots, X_n$ are independent and identically distributed with a common $\mathcal{N}(\mu + \mu', \sigma_0^2)$ distribution. Let the parameter space Θ be defined by

$$\Theta \equiv \{(\mu, \mu') : \mu \in \mathbb{R}, \mu' \in \mathbb{R}\}.$$

Finally, let model $\mathcal{P}$ be defined by

$$\mathcal{P} \equiv \{P_{\mu, \mu'} : (\mu, \mu') \in \Theta\}.$$

$\mathcal{P}$ is parametric.

(d) Let sample space Ω be defined by

$$\Omega \equiv \{(N_1, \ldots, N_n) : N_i \text{ is a nest}\}.$$

Let $\mathbf{X} : \Omega \to \mathbb{R}^n$ be the random vector defined by

$$\mathbf{X} \equiv (X_1, \ldots, X_n),$$
$$X_i(N_1, \ldots, N_n) \equiv \text{the number of eggs laid in } N_i.$$

Let $\mathbf{Y} : \Omega \to \mathbb{R}^n$ be the random vector defined by

$$\mathbf{Y} \equiv (Y_1, \ldots, Y_n),$$
$$Y_i(N_1, \ldots, N_n) \equiv \text{the number of eggs hatching in } N_i.$$

We assume that $(\mathbf{X}, \mathbf{Y}) \sim P_{\lambda, p}$, where $P_{\lambda, p}$ is a probability distribution on $\mathbb{R}^n \times \mathbb{R}^n$ according to which $(X_1, Y_1), \ldots, (X_n, Y_n)$ are independent and identically distributed with a common distribution function $F_{\lambda, p}$, where

$$F_{\lambda, p}(x, y) = \mathcal{P}(x; \lambda) \cdot \mathcal{B}(y; x, p)$$

Let the parameter space Θ be defined by

$$\Theta \equiv \{(\lambda, p) : \lambda > 0, p \in [0, 1]\}.$$

Finally, let model $\mathcal{P}$ be defined by

$$\mathcal{P} \equiv \{P_{\lambda,p} : (\lambda, p) \in \Theta\}.$$

$\mathcal{P}$ is parametric.

□

1.2. Problem 1.1.4: Stochastically Larger Distributions in Two-Sample Models

(a) Let U be any random variable and V be any other nonnegative random variable. Show that

$$F_{U+V}(t) \leqslant F_U(t) \text{ for every } t.$$

(If F_X and F_Y are distribution functions such that $F_X(t) \leqslant F_Y(t)$ for every t, then X is said to be stochastically larger than Y.)

(b) As in Problem 1.1.1 describe formally the following model. Two groups of n_1 and n_2 individuals, respectively, are sampled at random from a very large population. Each member of the second (treatment) group is administered the same dose of a certain drug believed to lower blood pressure and the blood pressure is measured after 1 hour. Each member of the first (control) group is administered an equal dose of a placebo and then has the blood pressure measured after 1 hour. It is known that the drug either has no effect or lowers blood pressure, but the distribution of blood pressure in the population sampled before and after administration of the drug is quite unknown.

Proof. (a) By the definition of distribution functions,

$$F_{U+V}(t) = P(U + V \leqslant t) = P(\{\omega : U(\omega) + V(\omega) \leqslant t\}),$$
$$F_U(t) = P(U \leqslant t) = P(\{\omega : U(\omega) \leqslant t\}).$$

Since V is nonnegative,

$$U(\omega) \leqslant U(\omega) + V(\omega).$$

Hence

$$\{\omega : U(\omega) + V(\omega) \leqslant t\} \subset \{\omega : U(\omega) \leqslant t\}.$$

Theorem 1.1.1 (i) of [**Dur19**] then implies

$$P(\{\omega : U(\omega) + V(\omega) \leqslant t\}) \leqslant P(\{\omega : U(\omega) \leqslant t\}).$$

(b) Let sample spaces Ω_1, Ω_2 be defined by

$$\Omega_1 \equiv \{(I_1, \ldots, I_{n_1}) : I_i \text{ denotes an individual}\}.$$
$$\Omega_2 \equiv \{(I_1, \ldots, I_{n_2}) : I_i \text{ denotes an individual}\}.$$

Let $\mathbf{X} : \Omega_1 \to \mathbb{R}^{n_1}$ be the random vector defined by

$$\mathbf{X} \equiv (X_1, \ldots, X_{n_1}),$$
$$X_i(I_1, \ldots, I_{n_1}) \equiv \text{the blood pressure of } I_i \text{ measured after 1 hour of drug.}$$

Let $\mathbf{Y} : \Omega_2 \to \mathbb{R}^{n_2}$ be the random vector defined by

$$\mathbf{Y} \equiv (Y_1, \ldots, Y_{n_2}),$$
$$Y_i(I_1, \ldots, I_{n_2}) \equiv \text{the blood pressure of } I_i \text{ measured after 1 hour of placebo.}$$

We assume that $\mathbf{X} \sim P_1$, where P is a probability distribution on $\mathbb{R}^{n_1}$ according to which $X_1, \ldots, X_{n_1}$ are independent and identically distributed with a common distribution function F.

Also, we assume that $\mathbf{Y} \sim P_2$, where P is a probability distribution on $\mathbb{R}^{n_2}$ according to which $Y_1, \ldots, Y_{n_1}$ are independent and identically distributed with a common distribution function G.

Since the drug either has no effect or lowers blood pressure, F is stochastically larger than G.

□

1.3. Problem 1.1.7: Symmetric Distributions and their Properties

Show that $Y - c$ has the same distribution as $-Y + c$, if and only if, the density or frequency function p of Y satisfies $p(c + t) = p(c - t)$ for all t. Both Y and p are said to be symmetric about c.

Proof. Let F_Y, F_{Y-c} and F_{-Y+c} be the distribution functions of $Y, Y - c$ and $-Y + c$. Then, by definition and Theorem 1.2.1 of [**Dur19**],

$$\begin{aligned} F_Y(t) &= P(Y \leqslant t), \\ F_{Y-c}(t) &= P(Y - c \leqslant t) = P(Y \leqslant c + t) = F_Y(c + t), \\ F_{-Y+c}(t) &= P(-Y + c \leqslant t) = 1 - P(Y < c - t) = 1 - F_Y(c - t) + P(Y = c - t). \end{aligned}$$

(1) If Y has a frequency function p, then

$$F_Y(t) = \sum_{y=-\infty}^{t} p(y).$$

Hence

$$\begin{gathered} F_{Y-c}(t) = F_{-Y+c}(t) \\ \Updownarrow \\ F_Y(c + t) = 1 - F_Y(c - t) + P(Y = c - t) \\ \Updownarrow \\ \sum_{y=-\infty}^{c+t} p(y) = 1 - \sum_{y=-\infty}^{c-t} p(y) + p(c - t) \\ \Updownarrow \\ \sum_{y=-\infty}^{c+t} p(y) = \sum_{y=c-t}^{\infty} p(y) \\ \Updownarrow \\ \sum_{y=-\infty}^{t} p(c + y) = \sum_{y=-\infty}^{t} p(c - y) \\ \Updownarrow \\ p(c + t) = p(c - t). \end{gathered}$$

(2) If Y has a density function p, then

$$F_Y(t) = \int_{-\infty}^{t} p(y)dy.$$

Hence

$$
\begin{gathered}
F_{Y-c}(t) = F_{-Y+c}(t) \\
\Updownarrow \\
F_Y(c+t) = 1 - F_Y(c-t) + P(Y = c-t) \\
\Updownarrow \\
\int_{-\infty}^{c+t} p(y)dy = 1 - \int_{-\infty}^{c-t} p(y)dy + 0 \\
\Updownarrow \\
\int_{-\infty}^{c+t} p(y)dy = \int_{c-t}^{\infty} p(y)dy \\
\Updownarrow \\
\int_{-\infty}^{t} p(c+y)dy = \int_{-\infty}^{t} p(c-y)dy \\
\Updownarrow \\
p(c+t) = p(c-t).
\end{gathered}
$$

□

1.4. Problem 1.2.1: Merging Options

Consider a parameter space consisting of two points θ_1 and θ_2, and suppose that for given θ, an experiment leads to a random variable X whose frequency function $p(x|\theta)$ is given by

$\theta \backslash x$	0	1
θ_1	0.8	0.2
θ_2	0.4	0.6

Let π be the prior frequency function of θ defined by $\pi(\theta_1) = \frac{1}{2}, \pi(\theta_2) = \frac{1}{2}$.

(a) Find the posterior frequency function $\pi(\theta|x)$.
(b) Suppose $X_1, \ldots, X_n$ are independent with frequency functions $p(x|\theta)$. Find the posterior $\pi(\theta|x_1, \ldots, x_n)$. Observe that it depends only on $\sum_{i=1}^n x_i$.
(c) Same as (b) except use the prior $\pi_1(\theta_1) = .25, \pi_1(\theta_2) = .75$.
(d) Give the value of $P(\boldsymbol{\theta} = \theta_1 | \sum_{i=1}^n x_i = .5n)$ for the two priors π and π_1 when $n = 2$ and 100.

PROOF. (a) See below.

$$\begin{aligned}
\pi(\theta=\theta_1|X=0) &= \frac{p(X=0|\theta=\theta_1)\pi(\theta=\theta_1)}{p(X=0|\theta=\theta_1)\pi(\theta=\theta_1)+p(X=0|\theta=\theta_2)\pi(\theta=\theta_2)} \\
&= \frac{0.8\cdot 0.5}{0.8\cdot 0.5+0.4\cdot 0.5} \\
&= \frac{2}{3}, \\
\pi(\theta=\theta_2|X=0) &= \frac{p(X=0|\theta=\theta_2)\pi(\theta=\theta_2)}{p(X=0|\theta=\theta_1)\pi(\theta=\theta_1)+p(X=0|\theta=\theta_2)\pi(\theta=\theta_2)} \\
&= \frac{0.4\cdot 0.5}{0.8\cdot 0.5+0.4\cdot 0.5} \\
&= \frac{1}{3}, \\
\pi(\theta=\theta_1|X=1) &= \frac{p(X=1|\theta=\theta_1)\pi(\theta=\theta_1)}{p(X=1|\theta=\theta_1)\pi(\theta=\theta_1)+p(X=1|\theta=\theta_2)\pi(\theta=\theta_2)} \\
&= \frac{0.2\cdot 0.5}{0.2\cdot 0.5+0.6\cdot 0.5} \\
&= \frac{1}{4}, \\
\pi(\theta=\theta_2|X=1) &= \frac{p(X=1|\theta=\theta_2)\pi(\theta=\theta_2)}{p(X=1|\theta=\theta_1)\pi(\theta=\theta_1)+p(X=1|\theta=\theta_2)\pi(\theta=\theta_2)} \\
&= \frac{0.6\cdot 0.5}{0.2\cdot 0.5+0.6\cdot 0.5} \\
&= \frac{3}{4}.
\end{aligned}$$

(b) See below.

$$\begin{aligned}
\pi(\boldsymbol{\theta}=\theta_1|x_1,\ldots,x_n) &= \frac{p(x_1,\ldots,x_n|\theta=\theta_1)\pi(\theta=\theta_1)}{p(x_1,\ldots,x_n|\theta=\theta_1)\pi(\theta=\theta_1)+p(x_1,\ldots,x_n|\theta=\theta_2)\pi(\theta=\theta_2)} \\
&= \frac{B(\sum_{i=1}^n x_i;n,0.2)\cdot\frac{1}{2}}{B(\sum_{i=1}^n x_i;n,0.2)\cdot\frac{1}{2}+B(\sum_{i=1}^n x_i;n,0.6)\cdot\frac{1}{2}}, \\
\pi(\boldsymbol{\theta}=\theta_2|x_1,\ldots,x_n) &= \frac{p(x_1,\ldots,x_n|\theta=\theta_2)\pi(\theta=\theta_2)}{p(x_1,\ldots,x_n|\theta=\theta_1)\pi(\theta=\theta_1)+p(x_1,\ldots,x_n|\theta=\theta_2)\pi(\theta=\theta_2)} \\
&= \frac{B(\sum_{i=1}^n x_i;n,0.6)\cdot\frac{1}{2}}{B(\sum_{i=1}^n x_i;n,0.2)\cdot\frac{1}{2}+B(\sum_{i=1}^n x_i;n,0.6)\cdot\frac{1}{2}}.
\end{aligned}$$

(c) See below.

$$\begin{aligned}
\pi_1(\boldsymbol{\theta}=\theta_1|x_1,\ldots,x_n) &= \frac{p(x_1,\ldots,x_n|\theta=\theta_1)\pi_1(\theta=\theta_1)}{p(x_1,\ldots,x_n|\theta=\theta_1)\pi_1(\theta=\theta_1)+p(x_1,\ldots,x_n|\theta=\theta_2)\pi_1(\theta=\theta_2)} \\
&= \frac{B(\sum_{i=1}^n x_i;n,0.2)\cdot\frac{1}{4}}{B(\sum_{i=1}^n x_i;n,0.2)\cdot\frac{1}{4}+B(\sum_{i=1}^n x_i;n,0.6)\cdot\frac{3}{4}}, \\
\pi_1(\boldsymbol{\theta}=\theta_2|x_1,\ldots,x_n) &= \frac{p(x_1,\ldots,x_n|\theta=\theta_2)\pi_1(\theta=\theta_2)}{p(x_1,\ldots,x_n|\theta=\theta_1)\pi_1(\theta=\theta_1)+p(x_1,\ldots,x_n|\theta=\theta_2)\pi_1(\theta=\theta_2)} \\
&= \frac{B(\sum_{i=1}^n x_i;n,0.6)\cdot\frac{3}{4}}{B(\sum_{i=1}^n x_i;n,0.2)\cdot\frac{1}{4}+B(\sum_{i=1}^n x_i;n,0.6)\cdot\frac{3}{4}}.
\end{aligned}$$

(d) When using the prior π,

$$\begin{aligned}
\pi(\boldsymbol{\theta} = \theta_1 | \sum_{i=1}^{2} x_i = 1) &= \frac{B(1;2,0.2) \cdot \frac{1}{2}}{B(1;2,0.2) \cdot \frac{1}{2} + B(1;2,0.6) \cdot \frac{1}{2}} \\
&= \frac{\binom{2}{1}(0.2)^1(0.8)^1 \cdot \frac{1}{2}}{\binom{2}{1}(0.2)^1(0.8)^1 \cdot \frac{1}{2} + \binom{2}{1}(0.6)^1(0.4)^1 \cdot \frac{1}{2}} \\
&= \frac{2}{5}, \\
\pi(\boldsymbol{\theta} = \theta_1 | \sum_{i=1}^{100} x_i = 50) &= \frac{B(50;100,0.2) \cdot \frac{1}{2}}{B(50;100,0.2) \cdot \frac{1}{2} + B(50;100,0.6) \cdot \frac{1}{2}} \\
&= \frac{\binom{100}{50}(0.2)^{50}(0.8)^{50} \cdot \frac{1}{2}}{\binom{100}{50}(0.2)^{50}(0.8)^{50} \cdot \frac{1}{2} + \binom{100}{50}(0.6)^{50}(0.4)^{50} \cdot \frac{1}{2}} \\
&\approx 1.57 \times 10^{-9}.
\end{aligned}$$

When using the prior π_1,

$$\begin{aligned}
\pi_1(\boldsymbol{\theta} = \theta_1 | \sum_{i=1}^{2} x_i = 1) &= \frac{B(1;2,0.2) \cdot \frac{1}{4}}{B(1;2,0.2) \cdot \frac{1}{4} + B(1;2,0.6) \cdot \frac{3}{4}} \\
&= \frac{\binom{2}{1}(0.2)^1(0.8)^1 \cdot \frac{1}{4}}{\binom{2}{1}(0.2)^1(0.8)^1 \cdot \frac{1}{4} + \binom{2}{1}(0.6)^1(0.4)^1 \cdot \frac{3}{4}} \\
&= \frac{2}{11}, \\
\pi_1(\boldsymbol{\theta} = \theta_1 | \sum_{i=1}^{100} x_i = 50) &= \frac{B(50;100,0.2) \cdot \frac{1}{2}}{B(50;100,0.2) \cdot \frac{1}{4} + B(50;100,0.6) \cdot \frac{1}{4}} \\
&= \frac{\binom{100}{50}(0.2)^{50}(0.8)^{50} \cdot \frac{1}{4}}{\binom{100}{50}(0.2)^{50}(0.8)^{50} \cdot \frac{1}{4} + \binom{100}{50}(0.6)^{50}(0.4)^{50} \cdot \frac{3}{4}} \\
&\approx 5.23 \times 10^{-10}.
\end{aligned}$$

□

Bibliography

[Dur19] R. Durrett, *Probability: Theory and examples*, Cambridge Series in Statistical and Probabilistic Mathematics, Cambridge University Press, 2019.